

PRAVEEN KUMAR MURUGAIAH

Senior Data Engineer | Big Data, AI Infrastructure & Cloud Platforms

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SUMMARY

Senior Data Engineer with 6+ years architecting distributed data and ML infrastructure across Spark, Kafka, Delta Lake, and Databricks. Delivered \$1M+ in measurable infrastructure savings, cut pipeline latency up to 63%, and led cross-functional engineering teams shipping production ETL, streaming, and AI systems for enterprise and Fortune 500 clients.

Core Competencies: Distributed Systems & Streaming (Spark, Kafka, Delta Lake) • Cloud Data Platforms (AWS, Azure, GCP) • ETL/ELT & Data Modelling • ML Pipeline Engineering • Team Leadership

WORK EXPERIENCE

Sr. Data Engineer, Big Data & AI — Epam Systems Inc.

Nov 2025 – Present

United States

- Designed and deployed high-throughput data layers via Databricks, utilizing PySpark & Spark SQL to streamline ETL workflows, resulting in 22% faster data processing times and 38% faster ML model deployment.
- Led a 3-engineer team to architect an LLM-powered Spark optimization agent that parses job code, driver logs, and Spark metrics to generate actionable code- and AWS cluster-tuning recommendations.
- Shipped the agent as a client-facing product recognized with a company innovation award; deployed across enterprise accounts managing 500GB+ pipelines using CI/CD, cutting customer infrastructure costs by 23%.

Sr. Data Engineer, MLOps — Paychex Inc.

Nov 2024 – Oct 2025

United States

- Led a 2-engineer team migrating on-prem Oracle payroll systems to a Medallion architecture on Delta Lake, improving distributed processing performance 90% for 1.6M+ client records using Databricks and Azure Data Factory.
- Designed and deployed cloud ETL pipelines against banking/NBFI client SLAs to support real estate and insurance decisioning, driving 30% ROI within 3 months of launch.
- Built a sub-3-second streaming fraud-detection pipeline using time-windowed aggregation, watermarking, and late-data handling on Databricks, improving detection response time 80% and reducing false positives 40%.

Data Engineer — Cummins Inc.

May 2022 – Nov 2024

United States

- Architected partitioned, incremental batch/streaming pipelines with Databricks, Kafka, and Spark SQL processing 12M+ daily finance records across 40+ sources, improving fault tolerance 15%.
- Eliminated the small-file problem via Delta Lake OPTIMIZE and Z-ORDER indexing, raising pipeline reliability from 80% to 91% and query performance 35%.
- Led end-to-end ML pipeline design with real-time feature engineering for predictive models, achieving a 77% F1 score and cutting false alerts 12% through statistical validation.

AI Data Engineer, AI Research Center — IIT Madras

Jul 2020 – Jun 2021

India

- Migrated an on-premises data warehouse to AWS Redshift using Glue, Athena, and S3 with hash partitioning and compression, generating \$678K in annual infrastructure savings.
- Re-architected ML and ETL pipelines with standardized engineering practices, cutting processing time 30% and accelerating project delivery timelines.

Data Engineer — Vola Finance Inc.

Jun 2019 – Jun 2020

New York, United States

- Designed a STAR-schema AWS database with SCD Type 2 historical tracking and optimized indexing, improving query speed and storage efficiency 20% to support a 50% surge in transaction volume.
- Engineered ML feature pipelines with SageMaker and Spark, deploying scikit-learn models that improved loan-approval risk-assessment accuracy 89%.
- Spearheaded data harmonization, deduplication, and CDC processing for 120K+ daily transactions while resolving network I/O bottlenecks via dynamic autoscaling and repartitioning — cutting processing time 30%, network I/O 77%, and infrastructure cost \$300K annually.

TECHNICAL SKILLS

Big Data & Streaming: Apache Spark, Databricks, Delta Lake, PySpark, Unity Catalog, Kafka, Hive, Snowflake, Airflow
Languages & Data Modelling: Python, SQL, dbt, MongoDB, Neo4j,, Star/Snowflake Schema Design, SCD, OBT
Cloud Platforms: AWS (Redshift, Glue, Athena, S3, SageMaker), Azure (Data Factory, Storage Accounts, Runtime Integration), GCP
ML/AI & Analytics: LLM Agent Development, Claude, GenAI, Predictive Modelling, Feature Engineering
Visualization & BI: Power BI, Tableau
Certifications: Databricks Certified Data Engineer Professional, Databricks Certified Machine Learning Professional, Claude AI Certified, AWS Certified Solutions Architect (In Progress)

EDUCATION

University of Illinois Urbana-Champaign	<i>Aug 2021 – May 2023</i>
M.S., Advanced Analytics Engineering, Minor in CS & Statistics	GPA 4.0/4.0
Madras Institute of Technology, Anna University	<i>Aug 2015 – Jun 2019</i>
B.E., Minor in CS & Mathematics	GPA 3.9/4.0